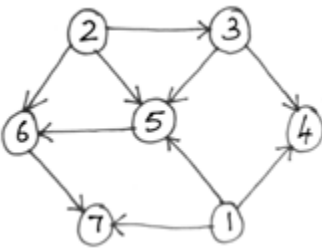
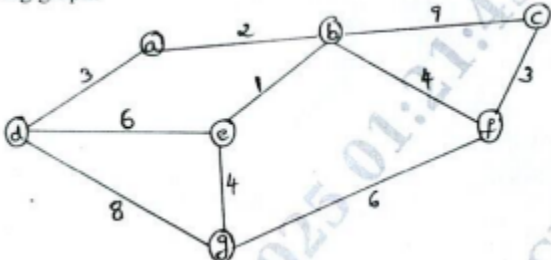
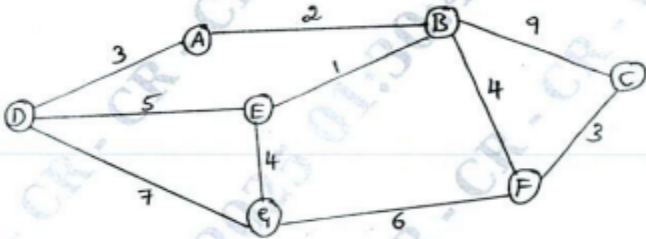




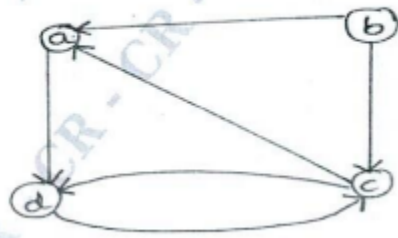
DEPARTMENT OF CSE
BCS401-ANALYSIS AND DESIGN OF ALGORITHMS
QUESTION BANK

MODULE - 01			
Q.No	QUESTION	RBT	CO
1	Write an algorithm to find uniqueness of elements in an array and give the mathematical analysis of this non recursive algorithm with steps	L2	CO1
2	Develop an algorithm to search an element in an array using sequential search. Calculate the best case, worst case and average case efficiency of this algorithm	L3	CO1
3	Explain asymptotic notations with example	L2	CO1
4	Give the general plan for analyzing the efficiency of the recursive algorithm	L2	CO1
5	Develop recursive algorithm for computing factorial of a positive number. Calculate the efficiency in terms of order of growth	L3	CO1
MODULE - 02			
1	What is divide and Conquer? Develop insertion sort algorithm on list of integers and find the time complexity.	L2	CO2
2	Obtain the topological sort for the following graph using DFS based algorithm 	L3	CO2

3	Apply merge sort for the following set of numbers 4, 9, 0, -1, 6, 8, 9, 2, 3, 12	L2	CO2
4	Sort the following keyword “ALGORITHM” by applying Quick Sort technique.	L2	CO2
5	Explain strassen’s matrix multiplication and derive its time complexity	L2	CO2
6	Explain the algorithm to implement insertion sort	L2	CO2
MODULE - 03			
1	Define AVL tree. Construct an AVL tree of the list of keys: 3, 6, 5, 1, 2, 4 indicating each step of key insertion and rotation.	L2	CO3
2	Consider the problem of searching for genes in DNA sequences using Horspool’s algorithm. A DNA sequence consists of a text on the alphabet {A, C, G, T} and the gene or gene segment is the pattern. a. Construct the shift table for the following gene segment of your chromosome 10: TCCTATTCTT b. Apply Horspool’s algorithm to locate the above pattern in the following DNA sequence: TTATAGATCTCGTATTCTTTTATAGATCTCCTATTCTT	L3	CO3
3	Define Heap. Constructing a heap for the list 1, 8, 6, 5, 3, 7, 4 by the bottom-up algorithm	L2	CO3
4	Construct a 2-3 tree for the list C, O, M, P, U, T, I, N, G. Use the alphabetical order of the letters and insert them successively starting with the empty tree.	L3	CO3
5	Build 2-3 tree for the list of keys 9, 5, 8, 3, 2, 4, 7 by indicating each step of key insertion and node splits.	L3	CO3
MODULE - 04			
1	Solve the all-pairs shortest path problem for the digraph with the following weight matrix:	L3	CO4

	$\begin{bmatrix} 0 & 2 & \infty & 1 & 8 \\ 6 & 0 & 3 & 2 & \infty \\ \infty & \infty & 0 & 4 & \infty \\ \infty & \infty & 2 & 0 & 3 \\ 3 & \infty & \infty & \infty & 0 \end{bmatrix}$														
2	Construct a Huffman Code. We are given the following characters and probabilities: <table border="1"><thead><tr><th>Character</th><th>A</th><th>B</th><th>C</th><th>D</th><th>–</th></tr></thead><tbody><tr><td>Probability</td><td>0.4</td><td>0.1</td><td>0.2</td><td>0.15</td><td>0.15</td></tr></tbody></table> Encode the text ABACABAD using the code	Character	A	B	C	D	–	Probability	0.4	0.1	0.2	0.15	0.15	L3	CO4
Character	A	B	C	D	–										
Probability	0.4	0.1	0.2	0.15	0.15										
3	Construct minimum cost spanning tree using Kruskals algorithm for the following graph. 	L3	CO4												
4	Write the Kruskal's algorithm to find the Minimum cost spanning tree of a given Weighted graph.	L2	CO4												
5	Find the Minimum Cost Spanning Tree using Prim's Algorithm	L2	CO4												
6	Apply Dijkstra's algorithm to find single source shortest path for the given graph by considering A as the source vertex. 	L3	CO4												
7	Define transitive closure of a graph. Apply Warshall's algorithm to compute	L3	CO4												

transitive closure of the following directed graph.



MODULE - 05

1	Differentiate Branch and Bound technique and backtracking	L2	CO5															
2	Apply backtracking to solve the following instance of subset-sum problem $S = \{3, 5, 6, 7\}$ and $d=15$. Construct a state space tree.	L3	CO5															
3.	Explain Greedy approximation algorithm to solve discrete knapsack problem	L2	CO5															
4	Explain n-Queens's problem with example using backtracking approach	L2	CO5															
5	Solve the following instance of knapsack problem by the branch-and-bound algorithm. Construct state space tree. The capacity of the knapsack $W=10$. <table border="1"><thead><tr><th>Item</th><th>Weight</th><th>Value</th></tr></thead><tbody><tr><td>1</td><td>4</td><td>\$ 40</td></tr><tr><td>2</td><td>7</td><td>\$ 42</td></tr><tr><td>3</td><td>5</td><td>\$ 25</td></tr><tr><td>4</td><td>3</td><td>\$ 12</td></tr></tbody></table>	Item	Weight	Value	1	4	\$ 40	2	7	\$ 42	3	5	\$ 25	4	3	\$ 12	L3	CO5
Item	Weight	Value																
1	4	\$ 40																
2	7	\$ 42																
3	5	\$ 25																
4	3	\$ 12																
6	Apply branch and bound approach to solve the instance of 0/1 knapsack problem. The capacity of the knapsack $W=10$. <table border="1"><thead><tr><th>Items</th><th>1</th><th>2</th><th>3</th><th>4</th></tr></thead><tbody><tr><td>Weight</td><td>4</td><td>7</td><td>5</td><td>3</td></tr><tr><td>Value</td><td>\$ 40</td><td>\$ 42</td><td>\$ 25</td><td>\$ 12</td></tr></tbody></table>	Items	1	2	3	4	Weight	4	7	5	3	Value	\$ 40	\$ 42	\$ 25	\$ 12	L3	CO5
Items	1	2	3	4														
Weight	4	7	5	3														
Value	\$ 40	\$ 42	\$ 25	\$ 12														
7	Explain the class of NP-Hard and NP-complete	L2	CO5															
8	Explain the following concepts: a. Tractable and intractable problems b. P Problems c. Non deterministic algorithm d. NP Problem	L1	CO5															

	e. NP complete problem		
9	What is backtracking? List out two advantages of backtracking strategy. Considering 4-Queens problem, provide two possible solutions to this problem using backtracking technique	L3	CO5
10	Apply back tracking method to solve sum of subset problem for the instance $d = 30$, $S = \{5,10,12,13,15,18\}$. Give all possible solutions with state space for construction	L3	CO5